ARQO: A Walk-Forward Study and Credit-Risk Framework for Prediction-Market Trader Lending

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Abstract—Prediction-market traders can realize large gains while also facing substantial loss risk. A historical trading record may describe activity and performance, but does not establish a borrower's probability of default, available collateral, or ability to repay. This paper presents ARQO , an engineering prototype that separates public trader-performance research from a bonded lending mechanism and an explicit portfolio credit-risk stress engine. The benchmark replays a sampled Polymarket dataset from November 2025 through October 2026: 764 of 768 scheduled markets, 3,647,433 fill rows, 3,906 wallet histories, 1,025,654 daily observations, and 15 of 15 checked month-boundary blocks. Across 11 walk-forward rounds, a consistency policy selected 220 wallet-month observations, of which 70.0% had positive next-month realized profit and loss (P&L); a seeded sample baseline had 27.7%. This descriptive difference is not a repayment result or causal estimate. The engine calculates exposure at default, bond-adjusted loss, expected loss, a beta-binomial correlated-default distribution, value at risk, expected shortfall, stress capital, and scenario pricing. Because ARQO has no mature repayment/default labels or funded shared reserve, default-probability inputs remain assumptions and pooled approval is hard-gated to rejection. This paper documents the prototype, benchmark, equations, results, and empirical work required before lending claims can be made.

Index Terms—prediction markets, credit risk, walk-forward evaluation, expected shortfall, beta-binomial, decentralized finance

I. Introduction

Prediction-market positions are risky claims whose value depends on event resolution. A trader's realized history can contain information about activity and outcomes, but it is not equivalent to complete wealth, return on equity, a probability forecast, or debt repayment capacity. This distinction is central to lending against trader performance: favorable history cannot be treated as collateral or as an enforceable future-income stream.

ARQO is a prototype for testing a narrower proposition: point-in-time prediction-market activity can be used to compare trader-selection policies, while credit exposure is evaluated independently under explicit loss assumptions. The application combines (i) a sampled monthly walk-forward benchmark, (ii) a bond-backed borrower request and separate lender funding flow, and (iii) a portfolio loss engine that exposes default and dependence scenarios. The benchmark does not calibrate the credit model.

The contributions are:

1) A reproducible Polymarket wallet and resolved-market sample with recorded capture gaps and Polygon month-boundary checks.
2) A walk-forward comparison of a consistency policy, monthly-P&L ranking, large-settled-buy cohorts, and a seeded sample baseline.
3) An integer-micro-USDC risk engine with bond haircuts, a beta-binomial default-count model, discrete-tail value at risk (VaR) and expected shortfall (ES), scenario capital, and a binding data-and-capital approval gate.

The results are descriptive engineering evidence. Measured next-month trader P&L is not lender return, and this study makes no claim that the consistency policy predicts repayment.

II. Related Risk Concepts

Credit-risk systems distinguish probability of default (PD), loss given default (LGD), and exposure at default (EAD). Their combination motivates expected-loss accounting, but the inputs must come from loan outcomes rather than unrelated performance measures [2]. Proper scoring rules such as the Brier score evaluate probability forecasts paired with realized outcomes; realized P&L is not a probability forecast [3]. For portfolio tails, conditional value at risk / expected shortfall accounts for losses beyond a quantile [4]. Systematic dependence can make concentrated exposures behave differently from independent loans, motivating explicit common-factor or correlation stress [5].

ARQO applies these concepts as a transparent scenario model. Its current PD and pairwise default-correlation values are neither calibrated nor observed. The model exposes how assumptions affect loss and capital, and keeps lending approval closed until actual outcome data and validation are available.

III. Prototype Mechanism

A borrower submits one to four associated EVM addresses, reviews dated public evidence, selects fixed terms, and escrows a first-loss bond. ENS can help discover associated addresses; it does not prove control, so each additional controller provides scoped consent. An authenticated report binds the evidence digest and request terms. A distinct lender reviews the request, acknowledges possible principal loss, and funds it. Voluntary repayment

TABLE I
Benchmark data and capture coverage.

Capture measure	Observed	Coverage
Scheduled markets	764 / 768	99.48%
Wallet histories	3,906 / 3,906	100%
Fill rows	3,647,433	–
Daily P&L observations	1,025,654	–
Verified month boundaries	15 / 15	100%
Candidate wallet-month rows	6,155	–
Forward decision rounds	11	–

returns the bond. A matured default can transfer the bond to the lender and record the remaining principal shortfall.

The local demonstration uses $100 principal, $102 due, and a $25 bond. With full bond recovery, the lender can still lose $75 of principal after a default. The bond is the only pledged asset in this lending desk. The system does not custody, liquidate, or automatically sweep a trader's external market positions or future P&L. Test funds and local report forwarding are not production liquidity or a public Chainlink DON execution receipt.

IV. Dataset and Reproducibility

A. Sources and Sampling

The benchmark uses public Polymarket wallet-P&L and trade data, resolved market outcomes, and Polygon source-block checks [1]. The fixed schedule spans November 2025 through October 2026 and samples BTC and ETH 15-minute markets at predetermined UTC dates and hours. Candidate discovery uses activity and large-settled-buy samples with per-month caps. This is a sampled panel, not a venue-wide census.

The replay captured 764 of 768 scheduled markets (99.48%), 3,647,433 fill rows, and histories for 3,906 of 3,906 requested wallets, totaling 1,025,654 daily observations. All 15 recorded monthly boundary blocks were verified. Four scheduled market samples were unavailable and were not silently imputed. The report artifact, capture manifest, policy tables, and CSV data are bound by a SHA-256 dataset commitment. Reproduction uses pnpm underwriting:replay –benchmark against the captured source files.

Table I summarizes sample completeness. The monthly candidate intake is nonuniform: the replay observed 3,545–19,403 active candidate wallets per decision month, then capped the sampled universe near 500 before applying evidence-quality filters. These counts describe the collection pipeline, not the whole venue.

B. Observed Features and Limits

For each wallet, available history supplies realized market P&L, turnover, trade count, and observed drawdown within dated windows. Complete account equity, opening bankroll, all open positions, and timestamped trader probability forecasts are unavailable. Consequently, these histories cannot establish return on equity, forecast skill, borrower solvency, or loan repayment. A settled-buy multiple is an exploratory event-level observation, not evidence of repeated skill.

V. Walk-Forward Benchmark

A. Design

At each monthly decision cutoff, candidate policies use only evidence available through that cutoff. The next month is held forward and measured for realized trader P&L, activity, and whether P&L is positive. The final month has no completed forward outcome. Eleven rounds therefore produce wallet-month selection events; an address can appear in multiple months, so event counts are not unique independent borrowers.

The consistency policy scores recent active months, realized efficiency, and observed drawdown. Alternative cohorts rank by largest monthly realized P&L, select a settled buy with a 10× to below-100× gross payout multiple, or select a buy with at least a 100× multiple. The baseline uses a deterministic seeded order over the eligible sampled universe. Cohort sizes differ because high-multiple events are sparse.

For cohort c, the reported positive-frequency statistic is

$$\widehat{\pi}_c = \frac{\sum_{j=1}^{n_c} \mathbf{1}[P\&L_j^+ > 0]}{n_c}, \tag{1}$$

where each j is a selected wallet-month event with an observed following month. This is a descriptive hit rate, not a PD estimate. The replay also reports median and total following-month realized P&L; these are trader P&L observations, not loan cash flows.

B. Results

Table II reports the forward sample. The consistency policy had a 70.0% positive next-month rate, compared with 27.7% for the seeded baseline, a descriptive difference of 42.3 percentage points. Ranking by largest monthly P&L yielded 73.6%, 3.6 percentage points above the consistency policy; this simple comparator therefore performed better on this sample. The large-win cohorts were smaller and less stable: 160 observations for 10×–below-100× buys and only four observations for at-least-100× buys. Figure 1 compares positive-P&L rates; Fig. 2 shows monthly variation for the consistency and baseline cohorts.

The positive-frequency gap is a research observation, not a statistically independent or causal estimate. Repeated wallet selection, small monthly cohorts, discovery and survivorship bias, market-regime effects, and sampled coverage can affect it. The four-event largest-multiple cohort is especially underpowered. No confidence interval or significance claim is reported. A later study should use wallet-clustered and time-blocked uncertainty estimates, preregistered policy thresholds, and a frozen out-of-time holdout.

TABLE II
Walk-forward trader outcomes. Each row is a selection event, not a loan. Active and positive counts use the evaluated wallet-month denominator. Mean and total P&L are sensitive to large observations.

Selection policy	n	Active	Positive	Rate	Median P&L	Mean P&L	Total P&L
Consistency score	220	203 (92.3%)	154	70.0%	$2,848.78	$17,408.59	$3,829,888.98
Largest monthly P&L	220	195 (88.6%)	162	73.6%	$16,981.79	$45,601.84	$10,032,404.40
10× to below 100× buy	160	134 (83.8%)	55	34.4%	-$3.47	-$12,855.13	-$2,056,820.43
At least 100× buy	4	4 (100%)	2	50.0%	-$3,076.38	-$3,409.58	-$13,638.32
Seeded sample baseline	220	182 (82.7%)	61	27.7%	-$55.80	-$2,492.02	-$548,243.34

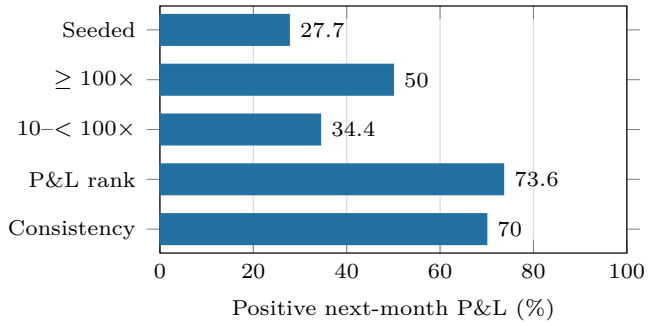


Fig. 1. Positive forward P&L frequency by policy. Sample sizes are 220, 220, 160, 4, and 220 wallet-month events, respectively.

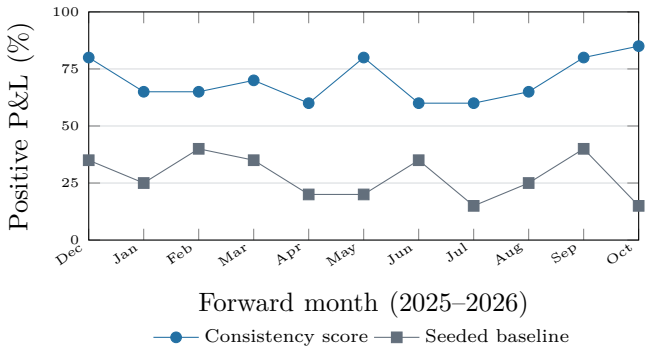


Fig. 2. Round-level positive next-month P&L rates. Each cohort contains 20 selected events per round; points are monthly descriptive rates, not independent replications.

VI. Credit-Risk Stress Engine

A. Exposure and Bond-Adjusted Loss

For loan i, let EAD_i be outstanding principal at default, B_i the posted bond, and $h \in [0,1]$ a stress recovery fraction. Recoverable bond and principal loss are

$$R_i = hB_i, \tag{2}$$

$$\ell_i = \max(0, EAD_i - R_i), \tag{3}$$

$$LGD_i = \begin{cases} \ell_i / EAD_i, & EAD_i > 0, \\ 0, & EAD_i = 0. \end{cases} \tag{4}$$

The prototype excludes unearned fee income from principal-loss calculations. With loan-level default probabilities p_i, expected loss is

$$EL = \sum_i p_i \ell_i = \sum_i p_i EAD_i LGD_i. \tag{5}$$

The application uses integer micro-USDC for money and integer basis points for ratios. It permits an explicit bond-recovery haircut stress; the default scenario assumes full recovery of the escrowed repayment token. This is a modeling assumption, not an empirical recovery estimate.

B. Correlated Default Count

For a homogeneous portfolio of N loans with marginal default probability p and pairwise correlation ρ between Bernoulli default indicators, ARQO models a shared latent default rate. For $0 < \rho < 1$,

$$\begin{aligned} \alpha &= p\frac{1-\rho}{\rho}, \\ \beta &= (1-p)\frac{1-\rho}{\rho}, \\ \Pr(K=k) &= \binom{N}{k}\frac{\mathrm{B}(k+\alpha, N-k+\beta)}{\mathrm{B}(\alpha,\beta)}, \\ k &= 0,\ldots,N. \end{aligned} \tag{6}$$

At $\rho = 0$, the implementation uses the binomial limit. At $\rho = 1$, it uses a common-shock mixture with mass $1-p$ at zero defaults and mass p at all defaults. For equal per-loan loss ℓ, portfolio loss is $L_k = k\ell$. The interface shows the full discrete distribution and cumulative probability.

This beta-binomial structure gives an interpretable one-factor stress, not a complete dependence model. Correlation is supplied as a scenario; it is not estimated from the trader benchmark or loan outcomes.

C. VaR, Expected Shortfall, and Capital

For confidence q, discrete value at risk is the smallest loss reaching cumulative probability q:

$$VaR_q = \inf\{x : \Pr(L \le x) \ge q\}. \tag{7}$$

Expected shortfall averages the worst $1-q$ probability mass:

$$ES_q = \frac{1}{1-q}\int_q^1 VaR_u \, du. \tag{8}$$

For a discrete distribution, the implementation includes only the fraction of the boundary probability atom needed

to complete the tail. This avoids including an entire atom when only part belongs to the worst tail.

The stress-capital policy is intentionally conservative:

$$K_{\text{req}} = \max(ES_q, N\ell), \qquad K_{\text{gap}} = \max(0, K_{\text{req}} - K_{\text{scenario}}). \tag{9}$$

The all-default term $N\ell$ imposes a common-shock loss floor. K_{scenario} is a user-controlled what-if amount and is never treated as funded capital by the approval gate.

D. Scenario Pricing

For fixed fee F, per-period capital charge r_K, and operating cost C, the model calculates

$$Revenue_{\text{fee}} = N(1-p)F, \tag{10}$$

$$Cost_{\text{capital}} = r_K \max(0, ES_q - EL), \tag{11}$$

$$Margin = Revenue_{\text{fee}} - EL - Cost_{\text{capital}} - NC. \tag{12}$$

Break-even PD is the greatest assumed p in the supported range for which modeled margin is nonnegative. This is a scenario threshold, not a borrower PD.

E. Binding Approval Gate

The decision function returns REJECT, REDUCE_LIMIT, or APPROVE. Approval requires mature repayment/default labels, PD calibrated from those labels, successful out-of-time validation, a nonempty requested cohort, and actual committed loss-absorbing capital. If evidence gates pass, the engine searches for the largest requested loan count whose capital requirement is covered and whose modeled margin is nonnegative. ARQO currently has zero mature loan labels, no calibrated PD, no passing out-of-time credit validation, and no funded shared reserve. The pooled decision is therefore REJECT, regardless of scenario slider settings.

F. Illustrative Scenario

At the interface defaults, a 20-loan cohort uses \$100 principal, a \$25 bond, a \$2 fee, 5% assumed PD, 25% pairwise default correlation, 100% bond recovery, 99% tail confidence, a 10% capital charge, zero operating cost, and zero scenario capital. Table III lists the inputs and model outputs; Fig. 3 compares loss and capital magnitudes. These values show how a small assumed PD can still produce a negative stress margin when the bond covers only one quarter of principal. They are not calibrated forecasts or historical lender results.

TABLE III
Illustrative 20-loan stress assumptions and outputs.

Input	Value	Output	Value
Loan count	20	Expected loss	\$75.00
Principal / loan	\$100	VaR (99%)	\$900.00
Bond / loan	\$25	Expected shortfall	\$1,043.36
Fee / loan	\$2	All-default loss	\$1,500.00
Assumed PD	5%	Required capital	\$1,500.00
Default correlation	25%	Scenario capital	\$0.00
Bond recovery	100%	Capital gap	\$1,500.00
Tail confidence	99%	Risk-adjusted margin	-\$133.84
Capital charge	10%	Break-even PD	0.28%

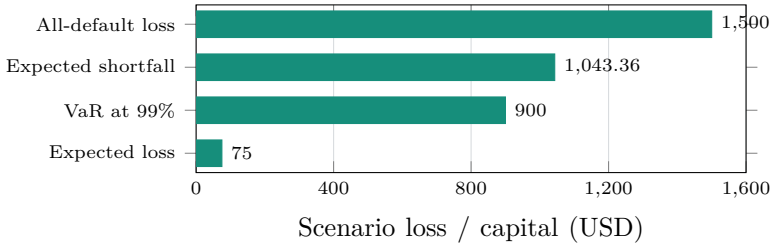


Fig. 3. Loss and capital levels under the stated scenario assumptions. Required capital is the maximum of expected shortfall and all-default loss; scenario capital is zero.

VII. Empirical Calibration Plan

The production model cannot be calibrated from the current market-performance dataset. A defensible study requires:

1) Point-in-time origination features, terms, and borrower consent recorded before funding.
2) Mature repayment outcomes that distinguish full repayment, partial repayment, default, cancellation, extension, and right-censoring.
3) Temporal train/validation/test splits and a final untouched out-of-time evaluation; random row splits risk leakage across repeated wallets and regimes.
4) PD calibration with uncertainty. A transparent Beta-Binomial baseline with prior $PD \sim \text{Beta}(a_0, b_0)$, d defaults among m matured loans, has posterior $\text{Beta}(a_0 + d, b_0 + m - d)$ and mean $(a_0+d)/(a_0+b_0+m)$. Limits should use conservative uncertainty-aware estimates, not a point score alone.
5) Separate discrimination and calibration reporting, including observed/expected defaults, reliability curves, Brier score and log loss on timestamped predictions, plus uncertainty by time and cohort.
6) Realized recovery after bond-transfer and servicing costs, dependence and concentration stress, utilization, and fixed/random benchmark policies.

Brier scoring is applicable only when a probability forecast exists before its outcome. ARQO does not currently collect those forecasts.

VIII. Discussion

The observed walk-forward results justify continued investigation of the selection policy, not a lending claim. A profitable next month does not imply that a trader will repay, that P&L is withdrawable, or that a wallet has no competing obligations. High performance frequencies also include large wallet-month outcomes, so user-level sample weighting and robust uncertainty analysis are essential before comparative claims.

The credit stress engine addresses a different question: what modeled principal loss and reserve requirement

follow from chosen PD, correlation, and recovery assumptions? Keeping this calculation separate from trader P&L prevents a proxy metric from being mislabeled as a loan outcome. The current rejection gate is a core safety property of the prototype, not a missing prediction.

IX. Limitations and Future Work

The benchmark is sampled and source-dependent; four planned market sessions are missing. Daily P&L does not contain total account wealth, opening bankroll, or timestamped forecasts. Wallets may recur across rounds, and selection may favor accounts visible in sampled markets. The data has no ARQO repayment/default labels, so empirical PD, lender returns, loss rates, model calibration, and loan profitability cannot be measured.

The homogeneous beta-binomial model omits heterogeneous PDs, dependence beyond a single pairwise parameter, dynamic wrong-way risk, stablecoin depeg, legal enforceability, liquidity, servicing-expense estimation, partial repayment, and recovery timing. Bond recovery assumes the configured token can be transferred as expected, except where a user applies a stress haircut. The local/testnet contract and report workflow have not been independently audited; no production pool or external-position custody is claimed.

Next work should add matured loan labels and realized recovery records; run wallet-clustered and time-blocked benchmark uncertainty; estimate and validate PD and dependence; stress reserve sufficiency and concentration; and complete independent contract/security review before production capital is accepted.

X. Conclusion

ARQO is an implemented research and testnet prototype for prediction-market trader credit. Its benchmark measures next-month trader outcomes across a sampled public dataset, while its risk engine models loan losses independently using explicit assumptions and a strict approval gate. Current data supports a descriptive observation that selected cohorts differed in forward P&L frequency; it does not support calibrated lending decisions. A future empirical paper can evaluate credit only after mature repayment outcomes, recovery data, temporal validation, and funded reserve evidence have been collected.

References

[1] Polymarket, Get a user's P&L series, Data API documentation. [Online]. Available: https://docs.polymarket.com/api-reference/wallet/get-a-users-pnl-series

[2] Basel Committee on Banking Supervision, CRE 30: Internal ratings-based approach, Basel Framework. [Online]. Available: https://www.bis.org/committees/bcbs/basel-framework/standard/cre.htm

[3] G. W. Brier, Verification of forecasts expressed in terms of probability, Monthly Weather Review, vol. 78, no. 1, pp. 1–3, 1950, doi: 10.1175/1520-0493(1950)078<0001:VOFEIT>2.0.CO;2.

[4] R. T. Rockafellar and S. Uryasev, Conditional value-at-risk for general loss distributions, Journal of Banking & Finance, vol. 26, no. 7, pp. 1443–1471, 2002, doi: 10.1016/S0378-4266(02)00271-6.

[5] M. B. Gordy, A risk-factor model foundation for ratings-based bank capital rules, Journal of Financial Intermediation, vol. 12, no. 3, pp. 199–232, 2003, doi: 10.1016/S1042-9573(03)00040-8.

[6] P. F. Christoffersen, Evaluating interval forecasts, International Economic Review, vol. 39, no. 4, pp. 841–862, 1998. [Online]. Available: https://users.ssc.wisc.edu/~bhansen/718/Christoffersen1998.pdf

[7] ENS, Resolution, ENS Documentation. [Online]. Available: https://docs.ens.domains/resolution/

[8] Chainlink, EVM read and write, CRE Documentation. [Online]. Available: https://docs.chain.link/cre/capabilities/evm-read-write